

# SECONDARY 1 FINAL EXAMINATION MATHEMATICS COMPULSORY PART

Time Allowed: 1 hour 30 minutes

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## INSTRUCTIONS

1. This paper consists of THREE sections: Section A (20 marks), Section B (41 marks), and Section C (8 marks).
2. **Section A** contains 20 multiple-choice questions. All questions carry equal marks (1 mark each).
3. **Section B** contains 7 compulsory long questions. Write your answers in the spaces provided.
4. **Section C** contains 3 challenging long questions. Answer **ONE** question only in Section C.
5. Unless otherwise specified, all working must be clearly shown.
6. Unless otherwise specified, numerical answers should either be exact or given to 3 significant figures.
7. The diagrams in this paper are not necessarily drawn to scale.

| Section      | Full Marks | Marks Obtained |
|--------------|------------|----------------|
| A            | 20         |                |
| B            | 41         |                |
| C            | 8          |                |
| <b>Total</b> | <b>69</b>  |                |

**SECTION A (20 marks)**

Answer ALL questions in this section. Choose the best answer for each question. Each question carries 1 mark.

1. Find the L.C.M. of  $2^2 \times 3 \times 5$ ,  $2 \times 3^2 \times 7$ , and  $3 \times 5^2$ .

A. 3  
B.  $2^2 \times 3^2 \times 5^2 \times 7$   
C.  $2 \times 3 \times 5$   
D.  $2^3 \times 3^3 \times 5^3 \times 7$

2. Evaluate  $1.8 \div \left(\frac{2}{5} + \frac{1}{2}\right) - 0.5$ .

A. 1.2  
B. 1.5  
C. 2.0  
D. 2.5

3. Evaluate  $(-2)^3 - (-5) \times (-3) + (-12) \div 4$ .

A. -26  
B. -20  
C. -10  
D. 4

4. If  $x = -3$  and  $y = -2$ , find the value of  $\frac{x^2 - 3y}{x - y}$ .

A. -15  
B. -3  
C. 3  
D. 15

5. Simplify  $(3a^2b^{-1})^2 \times (2ab^3)^{-1}$ .

A.  $\frac{9a^3}{2b^5}$   
B.  $\frac{18a^3}{b^5}$   
C.  $\frac{9a^3}{2b}$   
D.  $\frac{3a^3}{2b^5}$

6. Simplify  $\frac{4^{x+1} \times 2^x}{8^x}$ .
- A. 2  
B. 4  
C.  $2^{x+1}$   
D.  $4^{x+1}$
7. Solve the equation  $\frac{3x-1}{4} - \frac{1}{3} = \frac{5}{3}$ .
- A.  $x = 1$   
B.  $x = 2$   
C.  $x = 3$   
D.  $x = 7$
8. State the degree and the number of terms of the polynomial  $5x^3y^2 - 3x^2y + 8x - 12$ .
- A. Degree = 3, Terms = 4  
B. Degree = 5, Terms = 4  
C. Degree = 5, Terms = 3  
D. Degree = 6, Terms = 4
9. Arrange  $4x - 2 + 7x^3 - x^2$  in ascending powers of  $x$ .
- A.  $-2 + 4x - x^2 + 7x^3$   
B.  $7x^3 - x^2 + 4x - 2$   
C.  $-x^2 + 4x - 2 + 7x^3$   
D.  $-2 - x^2 + 4x + 7x^3$
10.  $200 \times (1 + 15\%) + 150 \times (1 - 20\%) =$
- A. 320  
B. 350  
C. 360  
D. 380
11. The price of a watch first increases by 20% and then decreases by 15%. What is the overall percentage change in price?
- A. Increased by 2%  
B. Increased by 5%  
C. Decreased by 2%  
D. Decreased by 5%

12. In a school club, 65% of the members are girls. If there are 28 boys in the club, how many members are there in total?

A. 52  
B. 70  
C. 80  
D. 90

13. Find the sum of interior angles of a hexagon.

A.  $180^\circ$   
B.  $360^\circ$   
C.  $560^\circ$   
D.  $720^\circ$

14. The stem-and-leaf diagram shows the test marks of 10 students:

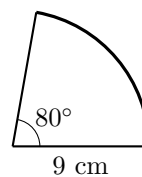
| Stem (10) | Leaf (1) |
|-----------|----------|
| 5         | 2 8      |
| 6         | 1 4 7    |
| 7         | 0 3 5    |
| 8         | 2 8      |

Find the mean mark of the students.

A. 68.0  
B. 69.0  
C. 70.0  
D. 71.0

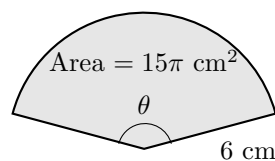
15. Find the arc length of a sector with radius 9 cm and central angle  $80^\circ$ .

A.  $2\pi$  cm  
B.  $4\pi$  cm  
C.  $6\pi$  cm  
D.  $8\pi$  cm



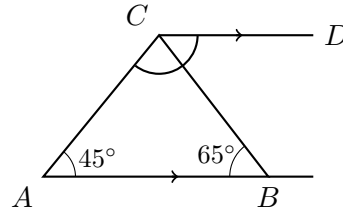
16. A sector has a radius of 6 cm and an area of  $15\pi$  cm<sup>2</sup>. Find its central angle.

A.  $100^\circ$   
B.  $120^\circ$   
C.  $150^\circ$   
D.  $160^\circ$



17. In  $\triangle ABC$ ,  $AB \parallel CD$ . If  $\angle BAC = 45^\circ$  and  $\angle ABC = 65^\circ$ , find  $\angle ACD$ .

- A.  $70^\circ$   
 B.  $110^\circ$   
 C.  $115^\circ$   
 D.  $135^\circ$



18. "Subtract 4 from twice  $x$ , then divide the result by the sum of  $y$  and 3." Express as an algebraic expression.

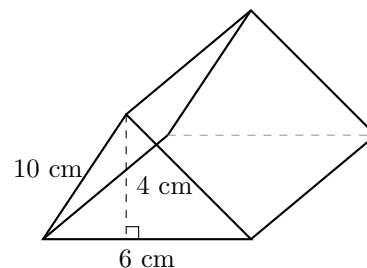
- A.  $\frac{2x - 4}{y + 3}$   
 B.  $\frac{2(x - 4)}{y + 3}$   
 C.  $\frac{4 - 2x}{y + 3}$   
 D.  $2x - \frac{4}{y} + 3$

19. A book costs  $\$x$ . A pencil costs  $\$y$ . If Sam buys 3 books and 5 pencils with a  $\$200$  note, how much change should he receive?

- A.  $\$(200 - 3x + 5y)$   
 B.  $\$(200 - 3x - 5y)$   
 C.  $\$(3x + 5y - 200)$   
 D.  $\$ \left( \frac{200}{3x + 5y} \right)$

20. A right prism has a triangular base with base length 6 cm and height 4 cm. If the height of the prism is 10 cm, find its volume.

- A.  $120 \text{ cm}^3$   
 B.  $240 \text{ cm}^3$   
 C.  $360 \text{ cm}^3$   
 D.  $480 \text{ cm}^3$



## SECTION B (41 marks)

Answer ALL questions in this section. Write your answers in the spaces provided.

**21.** Evaluate the following.

(a)  $\frac{3}{4} + \frac{5}{6} \times \frac{3}{10}$ . (2 marks)

(b)  $2.4 \div \left(1\frac{1}{2} - 0.3\right)$ . (2 marks)

(c)  $\frac{1}{1 + \frac{1}{1+\frac{1}{2}}}$ . (2 marks)

- 22.** The temperature in a freezer was  $-12^{\circ}\text{C}$  at 8:00 a.m. It rose by  $5^{\circ}\text{C}$  at noon and then dropped by  $9^{\circ}\text{C}$  at midnight.
- (a) Write down the changes in temperature using directed numbers. (2 marks)
- (b) Find the final temperature at midnight. (2 marks)
- (c) It is recommended to freeze meat at  $-30^{\circ}\text{C}$ . Find the temperature change needed. (2 marks)

(a)  $(x^3y^{-2})^3$  (2 marks)

(c)  $\left(\frac{2m^2}{n^{-1}}\right)^3 \div (4m^4n^2)$  (2 marks)



**24.** Solve the following linear equations in one unknown.

(a)  $3(2x - 5) = 4x + 1$ . (2 marks)

(b)  $\frac{x+3}{2} - \frac{2x-1}{5} = 2.$  (2 marks)



**26.** The cost of a handbag is \$500. The marked price is greater than the cost by 40%.

(a) Find the marked price of the handbag.

(2 marks)

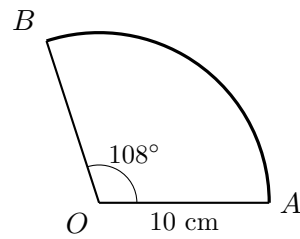
(b) During a promotional sale, the handbag is sold at a discount of 20%.

(i) Find the selling price of the handbag.

(ii) Find the profit or loss percentage.

(4 marks)

- 27.** The figure shows a sector  $OAB$  with radius  $r = 10$  cm and central angle  $\angle AOB = 108^\circ$ .

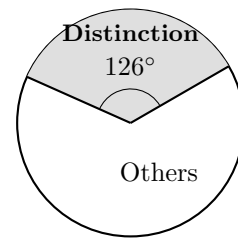


- (a) Find the arc length  $AB$  in terms of  $\pi$ . (2 marks)
- (b) Find the perimeter of the sector  $OAB$  correct to 3 significant figures. (2 marks)

**28.** The following figures show the mathematics test marks of two classes:

| Stem (10) | Leaf (1)  |
|-----------|-----------|
| 4         | 2 5 8     |
| 5         | 1 3 6 9   |
| 6         | 0 2 4 7 8 |
| 7         | 1 5 6     |
| 8         | 0 3 7     |
| 9         | 2 5       |

**Figure 1: Class 1A**



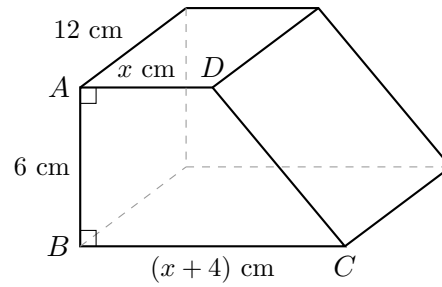
**Figure 2: Class 1B**

- Students scoring 70 or above are awarded a "Pass with Distinction". Find the percentage of students in Class 1A who obtained a "Pass with Distinction". (2 marks)
- Which class has a higher proportion of students receiving "Pass with Distinction"? Explain your answer. (4 marks)





- 30.** A right prism has a trapezoidal base  $ABCD$  where  $AD \parallel BC$ ,  $AB \perp BC$ ,  $AD = x$  cm,  $BC = (x + 4)$  cm, and  $AB = 6$  cm. The height of the prism is 12 cm.



- Express the base area of the prism in terms of  $x$ . (2 marks)
- If the total volume of the prism is  $432 \text{ cm}^3$ , form an equation in  $x$  and solve for  $x$ . (3 marks)
- A triangular prism is cut from the right prism above to form a cuboid. Find the maximum volume of that cuboid. (3 marks)





(4 marks)

(4 marks)

(4 marks)

– End of Paper –